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Detecting language through compression

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Problem: Can we measure shared information using compression?

AIT principles, implemented through compression, are used in a variety of practical applications: rediscover language genealogy without knowing linguistics, attributing authorship without knowing authors' styles, determining concept similarity without any knowledge of semantics, finding out Coronavirus-19 phylogeny while ignoring biological knowledge, and more. We explore some of these techniques now.

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Let's consider a set of reference texts T_i written in different languages. Such texts can be found in texts.zip. They have been stored in utf-8 format.
Then you may build a small text X (typically one-page long) in whatever language you want. This text will be used as a probe. You may cut-and-paste some text into your editor before saving it as X (instead of saving it directly from the browser) to eliminate html tags and links. Be sure it's saved in utf-8 format.

We want to see if **joint compression** with the T_i s can identify X 's language. To do so, we rank the T_i by increasing values of:
 $\text{length}(Z(T_i + X)) - \text{length}(Z(T_i))$

where Z is your favourite compressor. It would be great if the T_i that comes first correctly identifies the language (or the author) of X .

Use the program `TextDistance.py` to compute $\text{length}(Z(T_i + YourText)) - \text{length}(Z(T_i))$, where the T_i are the texts in various languages (make sure that $YourText$ is small as compared to T_i). Note: this program imports `NumberCompression.py`.



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